CLASS 04 / SEP 15 / U03

DESCRIPTIVE GEOMETRY

P3 BEGINS — WHAT A SOLID IS MADE OF, UNDERNEATH

P2a DUE

P3a ISSUED TODAY

POINT → LINE → EDGE → FACE → SOLID → BOOLEAN OPERATION

Six steps. Each one is built from the step before it.

POINT	LINE	EDGE	FACE	SOLID	BOOLEAN
three numbers `[x, y, z]`	two positions `[p1, p2]`	two vertices `[v1, v2]`	ordered corners `[v1, v2, v3]`	closed faces `[f1, f2, f3…]`	two solids `A − B`

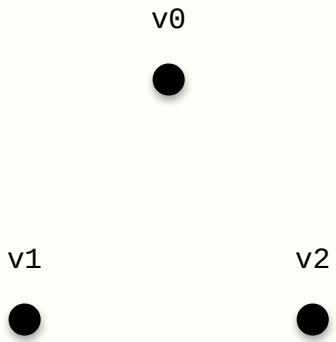
A POSITION IS NOT A DISPLACEMENT. THE SAME THREE NUMBERS MEAN DIFFERENT THINGS DEPENDING ON WHICH ONE YOU MEANT.

Nothing here is new geometry. It is the same collection, read at a different level: a face is a list of vertices, a solid is a closed list of faces, and in Week 6 an array will be a list of copies.

A FACE IS AN ORDERED LIST OF CORNERS

Same three vertices, two different faces. The order is the difference.

WINDING ORDER



f = [v0, v1, v2]

normal → toward you

f = [v0, v2, v1]

normal → away

WHY IT MATTERS LATER

A flipped face tells a renderer the inside is the outside.

It tells a Boolean operation it cannot decide what to keep.

Most broken renders and failed cuts this semester will start here.

CLOSED MEANS EVERY EDGE IS SHARED BY EXACTLY TWO FACES. CHECK IT BEFORE YOU BELIEVE IT.

THE PYRAMID

Five vertices, eight edges, five faces. Built by hand, driven by three numbers.

WHAT YOU ASSEMBLE

5

VERTICES

four base corners and an apex

8

EDGES

four around the base, four to the apex

5

FACES

four triangular sides and the base

EXPOSED INPUTS

`Width`

`Depth`

`Height`

Three numbers, and the whole solid rebuilds. That is the entire point of the exercise.

DO NOT FORGET THE BASE FACE. A PYRAMID WITH FOUR SIDES AND NO BOTTOM IS NOT A SOLID.

THEN IT GOES INTO A BOOLEAN OPERATION

Cube minus pyramid. The same subtraction as in P2, with a solid you built point by point.

01 BUILD IT	02 SUBTRACT IT	03 DRIVE IT	04 EXPLAIN IT
Five vertices, five faces, closed, with Width / Depth / Height exposed.	Cube minus pyramid. Name which solid is kept and which is removed.	Change the three inputs and show at least three settings of the result.	Face order, normals, closure — why it cuts cleanly instead of failing.

AN OPEN MESH HAS NO INSIDE. A BOOLEAN OPERATION NEEDS ONE.

The pyramid is the warm-up. Next week the same two lists describe your own module — the one you light and render for P3a, and array for P3b.

THE SEQUENCE

What we build together, in order, this class.

01	Point — locate a point with XYZ values; distinguish position from displacement.
02	Line — connect two positions with a curve.
03	Edge — connect two mesh vertices.
04	Face — order its corners and explain the normal.
05	Solid — build a pyramid from four base corners and an apex. Add four side faces and a base; check closure.
06	Boolean — subtract the pyramid from a cube, revisiting CSG.

THE SAME THING, WRITTEN OUT

A mesh is two lists. That is the entire idea, and it fits on one slide.

```
verts = [[-w, -d, 0],              # 0
         [ w, -d, 0],              # 1
         [ w,  d, 0],              # 2
         [-w,  d, 0],              # 3
         [ 0,  0, h]]              # 4  apex

faces = [[0, 1, 4], [1, 2, 4],
         [2, 3, 4], [3, 0, 4],
         [0, 3, 2, 1]]             # the base — do not forget it

pyramid = mesh(verts, faces)
result  = difference(cube, pyramid)
```

WHAT IT MAPS ONTO

point	edge
vector	face
parameter	mesh
vertex	solid
line	normal

verts IS A LIST OF POSITIONS. faces IS A LIST OF INDICES INTO IT.

ISSUED TODAY

P3a — DESCRIPTIVE GEOMETRY MODULE — MATERIAL AND LIGHTING

DUE WEEK 6 — Sep 29 / 7%

Find a simple panel in a built precedent and rebuild it from two lists — points and faces — driven by parameters. Present it on one sheet: one studio render, one material and a deliberate lighting setup.

FIND	DECIDE	BUILD	PRESENT
A PANEL	**WHAT MOVES**	**FROM TWO LISTS**	**ONE RENDER**
One precedent, simple. It has to tile 2 × 2, edge to edge.	Parameters first. One must change its shape, not its box.	Every point calculated from the parameters. Closed mesh.	Ground plane, one material, lighting chosen on purpose.

REQUIREMENTS — P3a

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01	Choose a simple panel from a built precedent that tiles edge to edge in a 2 × 2 grid. Credit the building, architect and source.
02	Define parameters first and explain each. Include a shape control such as aperture, fold depth or corner offset, beyond overall dimensions.
03	Build an editable, closed mesh from explicit point and face lists. Calculate point positions from the parameters.
04	State the ranges where the module closes and tiles. Diagram low, middle and high values of a shape parameter.
05	Use one 17 × 11 inch sheet with one studio render of the module on a ground plane — one material, a deliberate lighting setup (key, fill and rim, or one sun and sky) and no entourage. Record the material and light settings on the sheet.
06	Beside the render, show point and face lists, a parameter table and the vertex order of one face. Explain normals, closure and reversed face winding.

NOW — BUILD THE PYRAMID

Rest of the class: from a single point to a closed solid that cuts.

✓	One point placed from an XYZ vector, and you can say why it is a position
✓	Four base corners and an apex, with Width / Depth / Height exposed
✓	All five faces joined — four sides and the base — and closure checked
✓	Cube minus pyramid working, off the same three inputs
✓	P2a submitted, and your P3a precedent panel chosen

LEAVE WITH A SOLID THAT CUTS. THE SHEET IS THE EASY PART.

BEFORE NEXT CLASS

NEXT	Week 5 continues descriptive geometry. We move from the pyramid to your own module, then set up its material and lighting for the P3a sheet.
FIND	Your panel. One building, one repeating component, simple enough to write as two lists, and it has to tile 2 × 2. Credit the building, architect and source.
BUILD	Start your module: parameters written down first, then the point list. Bring the file — next class we close it, light it and render it.
DUE	P2a is due today. P3a, the module sheet: DUE WEEK 6 — Sep 29 / 7%.